

0.1 Section 2.1 - The Joy of Sets

Definition 1 (Sets and Elements)

A **set** is an unordered collection of objects called **elements**

$$\mathbb{N} = \{1, 2, 3, 4, \dots\} \quad (\text{Natural Numbers})$$

$$\text{Whole Numbers} = \{0, 1, 2, 3, \dots\}$$

$$\mathbb{Z} = \{0, -1, 1, -2, 2, -3, 3, \dots\}$$

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}$$

$$\mathbb{R} = \text{limit of all convergent sequences of rational numbers}$$

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$$

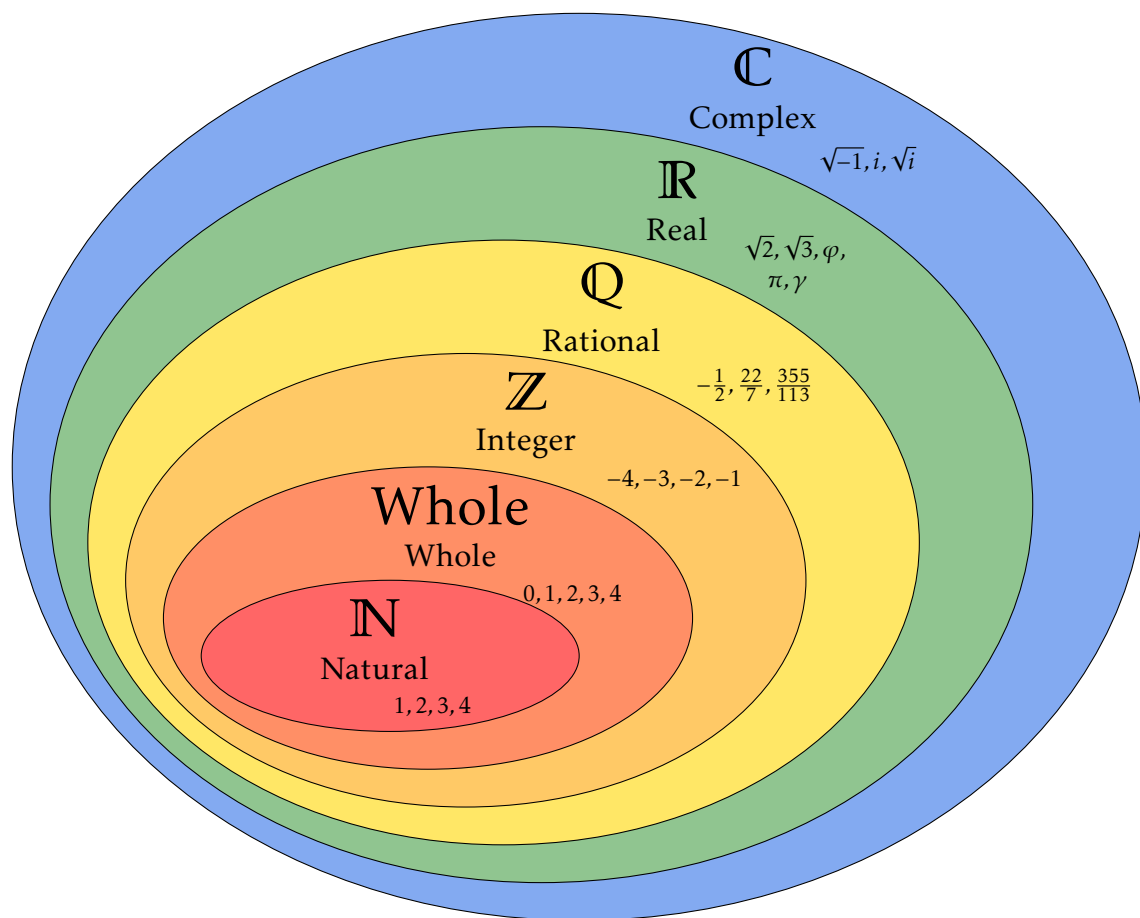
Note

This notation for $\mathbb{Q}$ is called **Set Builder Notation**

It's read as "Q is the set $\frac{a}{b}$ such that $a, b \in \mathbb{Z}, b \neq 0$ "

Note

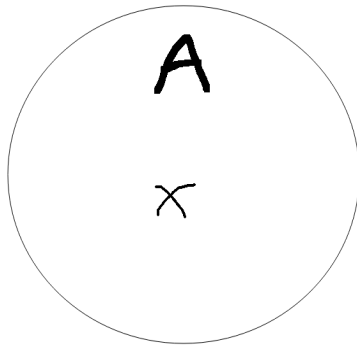
Notice that every natural number is a whole number. So we say "the set of natural numbers is a **subset** of the set of whole numbers"



Definition 2 (Elements in Sets)

If x is an element of the set A , then we write

$$x \in A$$



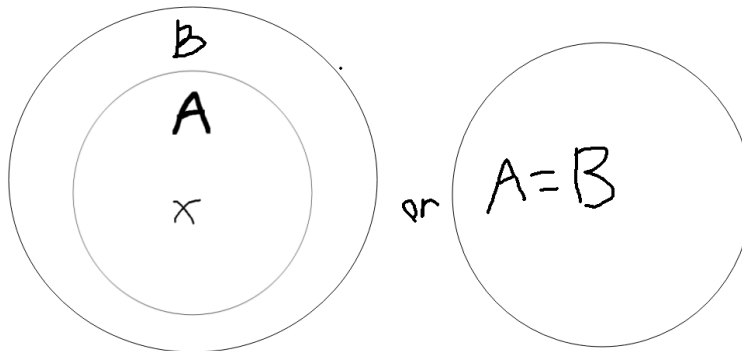
Definition 3 (Empty Set)

The set containing no elements is called the **empty set**, and is denoted by $\{\}$ or $\emptyset$

Given sets A and B such that for all $x \in A$ we have $x \in B$, then we say **A is a subset of B** and we write

$$A \subseteq B$$

Which can mean two things:



Definition 4 (Proper Sets)

If there exists $x \in B$ with $x \notin A$, then we say that A is a **proper subset** of B and we write

$$A \subsetneq B$$

Definition 5 (Cardinality)

Given a set A , the **cardinality** of a set is the number of elements in the set A , denoted by

$$|A|$$

Definition 6 (Power Set)

The **power set** of a given set S is the set of all subsets of S , denoted by

$$\mathcal{P}(S)$$

$$|\mathcal{P}(S)| = 2^{|S|}$$

Note

This includes the empty set, $\{\}$

Definition 7 (Cartesian Product)

Given sets A, B , the **cartesian product** is the set

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

$$|A \times B| = |A| \cdot |B|$$

$$A^2 = A \times A$$

Example 1

Let $A = \{1, 2, 3\}, B = \{1, 2\}$

(a) Is $A \subseteq B$?

Solution:

No since $3 \in A, 3 \notin B$

(b) Is $B \in A$?

Solution:

Yes since for all $x \in B, x \in A$

(c) What is the power set of B ?

Solution:

$$\mathcal{P} = \{\{\}, \{1\}, \{2\}, \{1, 2\}\}$$

(d) What is the power set of A ?

Solution:

$$\mathcal{P}(A) = \{\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Note

If A is a finite set (ie $|A|$ is finite), then $|\mathcal{P}(A)| = 2^{|A|}$

(e) Determine $A \times B$

Solution:

$$A \times B = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)\}$$

$$B \times A = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\}$$

$$A \times B \neq B \times A$$

$$|A \times B| = |B \times A| = |A| \cdot |B|$$